Open Data Framework

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Overview

Introduction

General Architecture

The Service Registry

Open Data Portal

Introduction

- Part of a Master's Thesis project
  - Implementing of Design Patterns in Jolie
- Intention that a Data Portal would benefit

The Problem

- Cooperation between three parties

The Problem

- Cooperation between three parties
  - ATKI
  - Odense Municipal
  - SDU

The Problem

- Cooperation between three parties
  - ATKI
  - Odense Municipal
  - SDU
- Allow receiving data from select sender
- Allow retrieval by anyone

General Architecture

- Load Balancer/Service Registry
- Services
- Potential Circuit Breakers

Service Registry/Load Balancer - General Description

- Service Registry
  - Store and Retrieve Binding Information
  - Fixed Location
  - Variations
- Load Balancer
  - Delegates Request
  - Enforcement Depends on Approach

Service Registry Variation

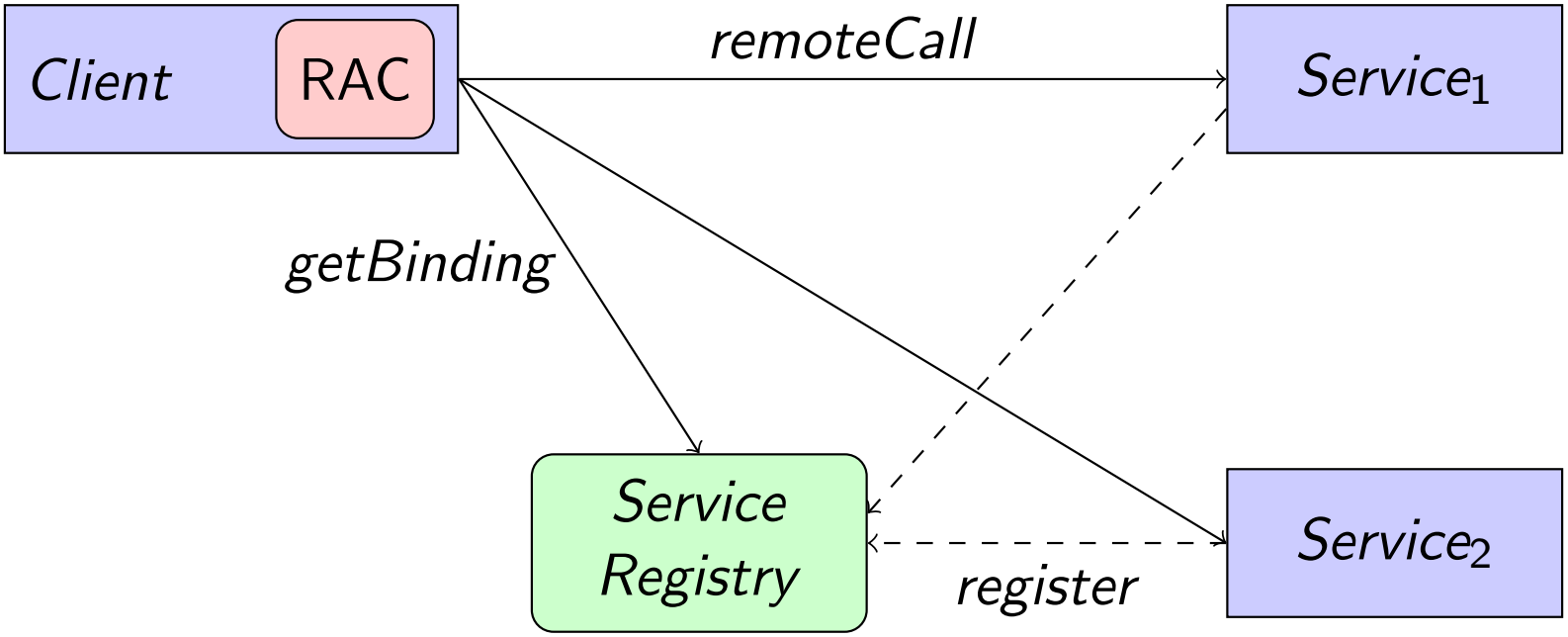

Figure 1: Service Registry with Registry Aware Client (RAC)

Service Registry Variation

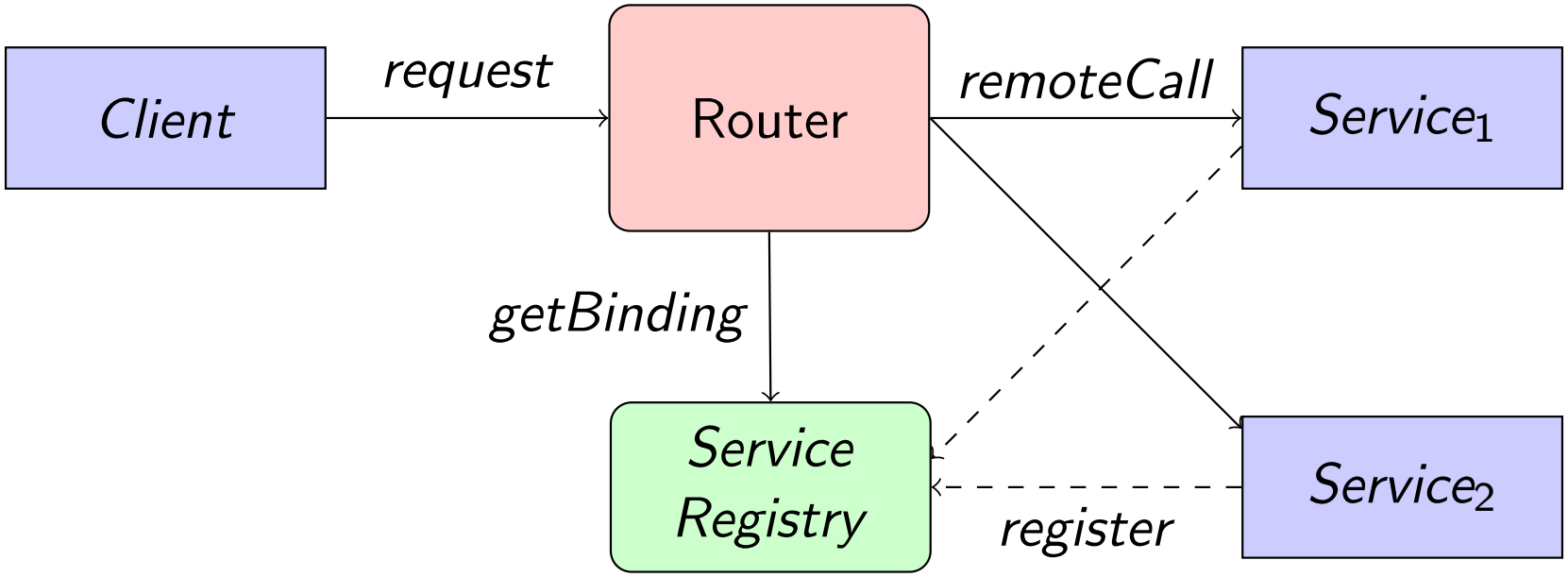

Figure 2: Service Registry with Service-side Service Discovery

SR - Design & Implementation

- Service Registry for Odense Kommune
- Router not viable
- Needed Operations
  - Register
  - Deregister
  - Query
  - Search/List
- Security
- Heartbeat

Service Registry Web Interface

Home About Setup SDU Search by name...

Servicename	Owner	Location	Protocol	Interface	Documentation
ParkingServiceWWW	Jonas Malte Hinchely	opendata.sdu.dk:8667	http	Download Interface	Show Documentation
LiveParkingData	Jonas Malte Hinchely	opendata.sdu.dk:8004	http	Download Interface	Show Documentation
ParkingService	Jonas Malte Hinchely	opendata.sdu.dk:8665	sodep	Download Interface	Show Documentation
TrafficData ATKI Service	Jonas Malte Hinchely	opendata.sdu.dk:8022	sodep	Download Interface	Show Documentation

Listing 1 Parking Service Registering

```
readFile@File( { .filename = "PService.iol" } )
             ( interfaceFile );
with ( registerRequest ) {
    .serviceName = PARKING_SERVICE_NAME;
    /* ... */
    .binding << {
        .location = PARKING_SERVICE_LOCATION,
        .protocol = PARKING_SERVICE_PROTOCOL_STRING
    }
};
registerService@RegisterService( registerRequest )
                                 ( success )
```

Listing 2 Registry Utilization

```
queryServices@ServiceRegistry( "ParkingService" )
                               ( res1 );
queryServices@ServiceRegistry( "LiveParkingData" )
                               ( res2 );
ParkingExtService.location = "socket://" + res1.Loc;
ParkingExtService.protocol = res1.Prot;
LiveParkingExtService.location = "socket://" + res2.Loc;
LiveParkingExtService.protocol = res2.Prot;
getLiveParkingData@LiveParkingExtService()( liveData );
get_info@ParkingExtService()( result );
```

Case Study - Description

- Data provided by Odense Kommune
  - Initially from 3rd party
- Traffic Data

The Data

- Station
- Station Data

Station Example

- status
- direction
- expectedInterval
- longitude
- stationType
- name
- installDate
- source
- stationID
- latitude

Station Example

- status - **Active**
- direction - **?**
- expectedInterval - **5**
- longitude - **10.5019**
- stationType - **OE Fartviser [...]**
- name - **Birkum Nord**
- installDate - **28-11-2011**
- source - **ATKI A/S**
- stationID - **51910323**
- latitude - **55.3417**

Station Data Example

- lane
- display
- flash
- stationID
- datetime
- length
- messGroup
- gap
- speed
- class
- wrong_dir

Station Data Example

- lane - **1**
- display - **0**
- flash - **0**
- stationID - **46171141**
- datetime - **2017-02-28 18:20:27**
- length - **42**
- messGroup - **4e5765d3-f670-435e-bd4f-9ff9cdc8dfd0**
- gap - **24240**
- speed - **42**
- class - **2**
- wrong_dir - **0**

The Method

- Defined and Implemented TrafficATKI-service
- Receives and allows for Retrieving Data
- HTTP

Receiving the Data

- JSON
- Python
- Two operations for receiving:
  1) receiveStation(StationRequest)(bool)

Receiving the Data

- JSON
- Python
- Two operations for receiving:
  1) receiveStation(StationRequest)(bool)
  2) receiveStationData(StationDataRequest)(bool)

```
header = { 'Content-Type' : "application/json" }
address = 'http://ServiceLocation:Port/receiveStation'
for i in stations:
    r = requests.post(address, headers=h, data=i)
```

Retrieving the Data

609587 rows (currently)

- getMessGroups(DateRange)(MessGroups),
- getByDate(DateRequest)(DateDataResponse),
- getStationDataCount(void) (long),
- getStationCount(void) (long),
- getStations(void)(StationResponse),
- getByMessGroup(string)(MessGroupDataResponse)

The Service

- TrafficATKI-service
- Database
- Circuit Breaker

Result

Result

- Worked as Intended

Result

- Worked as Intended
- Bloated Data Storage

Result

- Worked as Intended
- Bloated Data Storage
- Available data
- Received up until 5th of September

Roadmap/Future

- Generalized services

Roadmap/Future

- Generalized services
- Uploading and running services

Roadmap/Future

- Generalized services
- Uploading and running services
- Being used for Kolding Municipal